

Question 1. Extended Monty Hall Problem with Four Doors

A prize is hidden behind one of four doors, while the other three doors hide goats. Assume that the prize is equally likely to be behind any door. The game proceeds according to the following rules:

1. You first choose one door, but it remains closed.
2. The host knows where the prize is. From the three doors that you did not choose, the host opens two doors that both hide goats.
3. If the host has more than one possible pair of goat doors to open, the host selects one of the possible pairs at random.
4. Exactly two doors now remain closed: your original choice and one other door. You may either stay with your original choice or switch to the other unopened door.

Answer the following questions. Clearly state the events and conditional probabilities used in your reasoning.

- (a) Before the host opens any doors, what is the probability that your initial choice contains the prize?
- (b) What is the probability of winning if you always stay with your initial choice?
- (c) What is the probability of winning if you always switch to the only other unopened door?
- (d) Which strategy should you use? Explain why the host's action changes the information available to you but does not change whether your initial choice was correct.

Question 2. A Scalar Bayesian Analysis and the Kalman Gain

In data assimilation, a background estimate and an observation are combined to obtain an analysis estimate, which can then be used as the initial condition for numerical weather prediction. In the scalar case, suppose the observation operator is the identity, so that the background and observation represent the same physical quantity at the same location.

Let the background estimate be x_b and the observation be y . Assume that their errors are unbiased, mutually independent, and Gaussian, with variances σ_b^2 and σ_o^2 , respectively. The analysis is

$$x_a = x_b + K(y - x_b), \quad K = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_o^2}, \quad (1)$$

where K is the scalar Kalman gain. Under these assumptions, this expression is both the minimum-error-variance linear estimate and the maximum a posteriori (MAP) estimate.

Answer the following questions:

- (a) In a Bayesian formulation, write the prior density $p(x)$ implied by the background estimate and the likelihood $p(y | x)$ implied by the observation. Use Bayes' theorem to write the posterior density $p(x | y)$ up to a normalization constant. Identify which density is the prior and which is the posterior.
- (b) Show that maximizing $p(x | y)$ is equivalent to minimizing

$$J(x) = \frac{(x - x_b)^2}{2\sigma_b^2} + \frac{(y - x)^2}{2\sigma_o^2}. \quad (2)$$

Differentiate $J(x)$ and verify that its minimizer has the Kalman-gain form given above.

- (c) Given

$$x_b = 10^\circ\text{C}, \quad y = 15^\circ\text{C}, \quad \sigma_b^2 = 1^\circ\text{C}^2, \quad \sigma_o^2 = 0.5^\circ\text{C}^2,$$

calculate the Kalman gain and the analysis value that maximizes the posterior density. Briefly explain why the analysis lies closer to either the background or the observation.

Question 3. Bayesian Interpretation of a Heavy-Rain Warning

Suppose that, according to the local climatology, the probability of heavy rainfall on a randomly selected day is

$$P(H) = 0.10,$$

where H denotes the occurrence of heavy rainfall. A meteorological warning system has the following characteristics:

$$P(W | H) = 0.90, \quad P(W | H^c) = 0.20,$$

where W denotes that the system issues a heavy-rain warning, and H^c denotes that heavy rainfall does not occur. Thus, the system issues a warning on 90% of heavy-rain days and on 20% of non-heavy-rain days.

Answer the following questions:

1. Identify the prior probability of heavy rainfall, the likelihood of receiving a warning when heavy rainfall occurs, and the false-positive probability of the warning system.
2. Using the law of total probability, calculate the overall probability that the system issues a warning:

$$P(W) = P(W | H)P(H) + P(W | H^c)P(H^c).$$

3. Use Bayes' theorem to calculate the probability that heavy rainfall will occur given that a warning has been issued:

$$P(H | W) = \frac{P(W | H)P(H)}{P(W)}.$$

4. Consider 1000 days with the same statistical characteristics. Construct a table showing the expected numbers of:
 - heavy-rain days with a warning;
 - heavy-rain days without a warning;
 - non-heavy-rain days with a warning; and
 - non-heavy-rain days without a warning.

Use this table to verify your answer to Question 3.

5. Explain why $P(H | W)$ is considerably smaller than $P(W | H)$. What role does the climatological frequency of heavy rainfall play in interpreting the warning?
6. During the wet season, suppose the prior probability of heavy rainfall increases to $P(H) = 0.30$, while the characteristics of the warning system remain unchanged. Recalculate $P(H | W)$ and explain why the posterior probability changes even though the warning system itself has not changed.

Question 4. The Sleeping Beauty Paradox

Researchers put Sleeping Beauty to sleep on Sunday. They then toss a fair coin in another room.

- If **Heads**: She is woken up on Monday for an interview. Afterwards, she is given an amnesia-inducing drug and put back to sleep until the experiment ends.
- If **Tails**: She is woken up on Monday for an interview, given the amnesia drug, and put back to sleep. She is then woken up **again** on Tuesday for a second interview, and again given the drug.

During any interview, she has no idea what day it is, nor does she remember any previous awakenings. When she wakes up, the researchers ask her: “What is the probability that the coin landed Heads?”

(a) The 1/2 Position (No New Information)

From the perspective that “waking up” does not constitute new information:

- Use Bayes’ theorem to prove that her probability for Heads remains $1/2$. Clearly define your prior probabilities and the likelihood of the event “she wakes up” under each coin outcome.
- Briefly explain why the probability is not updated.

(b) The 1/3 Position (The Subjective Experience)

From the perspective of her subjective experience, prove that the probability of Heads should be updated to $1/3$. Provide the following two derivations:

- Suppose the entire experiment is repeated $2N$ times. Calculate the total number of “awakenings” she will experience, and how many of those awakenings occur when the coin is Heads.
- List all possible mutually exclusive states she could be in upon waking (i.e., combinations of the day and the coin outcome). Apply the principle of indifference to these states (e.g., if she is told it is Monday, Heads and Tails are equally likely; if she is told it is Tails, waking up on Monday and Tuesday are indistinguishable experiences) to determine the probability of Heads.